- A value of 2 indicates that additional interactions described in Section 5.7.3, "Custom Commissioning Flow" are required for initial device setup in order to be successfully commissioned by any Commissioner.

5.1.1.4. Discovery Capabilities Bitmask

An 8-bit capabilities bitmask SHALL be included in machine-readable formats.
Rationale: The Discovery Capabilities Bitmask (see Table 58, "Discovery Capabilities Bitmask") contains information about the device's available technologies for device discovery (see Section 5.4, "Device Discovery").

5.1.1.5. Discriminator value

A Discriminator SHALL be included as a 12-bit unsigned integer, which SHALL match the value which a device advertises during commissioning. Manufacturers SHALL NOT use a common discriminator value for all devices, to allow a commissioner to distinguish between multiple advertising devices. It is RECOMMENDED that manufacturers use sequential generation with rollover for devices that might be sold or purchased in a multi-pack in order to reduce the chance of discriminator collision between devices in the same pack.

For machine-readable formats, the full 12-bit Discriminator is used. For the Manual Pairing Code, only the upper 4 bits out of the 12-bit Discriminator are used.
Rationale: The Discriminator value helps to further identify potential devices during the setup process and helps to improve the speed and robustness of the setup experience for the user.

5.1.1.6. Passcode

A `Passcode` SHALL be included as a 27-bit unsigned integer, which serves as proof of possession during commissioning. The 27-bit unsigned integer encodes an 8-digit decimal numeric value, and therefore SHALL be restricted to the values `0x0000001` to `0x5F5E0FE` (`00000001` to `99999998` in decimal), excluding the invalid Passcode values.

Rationale: The Passcode establishes proof of possession and is also used as the shared secret in setting up the initial secure channel over which further commissioning steps take place.

5.1.1.7. TLV Data

Variable-length TLV data using the TLV format MAY be included in machine-readable formats providing optional information. More details about the TLV can be found in Section 5.1.5, "TLV Content".

5.1.2. Onboarding Material Representation

In order for the users of Matter products to recognize the onboarding material, and be able to use it easily, it is important to keep the representations of the onboarding material unified and of certain minimum size. To support this the Matter Brand Guidelines specify the characteristics like composition, colors, font, font size, QR Code size and digit-grouping of the Manual Pairing code.

When the onboarding material is printed on product or packaging material it SHALL follow the Matter Brand Guidelines.

